

Naive Bayes Classifier | Part 3: Mutually Exclusive Events

1 What are Mutually Exclusive Events?

- Two events are **mutually exclusive** if **they cannot happen at the same time**.
If one event occurs, the other is impossible.

Example 1 (Dice):

- Event A: Rolling a 2.
- Event B: Rolling a 5.
- You cannot roll both 2 and 5 in a single dice throw. → Mutually exclusive.

Example 2 (Weather):

- Event A: It is raining.
- Event B: It is snowing (assuming these cannot happen together in your region).
- If one happens, the other cannot.

2 Probability Rule for Mutually Exclusive Events

If A and B are mutually exclusive:

$$P(A \cap B) = 0$$

And the probability of **A OR B** happening is:

$$P(A \cup B) = P(A) + P(B)$$

👉 Example:

- Event A: Dice shows 3 → $P(A) = \frac{1}{6}$.
- Event B: Dice shows 6 → $P(B) = \frac{1}{6}$.
- Since they're mutually exclusive:

$$P(A \cup B) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

3 Difference from Independent Events

- Independent events:** Can happen together, but one does not affect the probability of the other.
- Mutually exclusive events:** Cannot happen together at all.

Example:

- Tossing two coins: (Head on first, Tail on second) → **independent**.
 - Rolling a dice: (3 and 6 at the same time) → **mutually exclusive**.
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4 Why learn this for Naive Bayes?

Understanding **independence vs mutual exclusivity** is important in probability:

- Naive Bayes relies on the **independence assumption** (features independent given class).
- Mutually exclusive events remind us that some events/features cannot co-exist, so in those cases probabilities must be handled differently.

For Example:

- A patient **cannot** be both "Male" and "Female" in a dataset → those are mutually exclusive categories.
 - But a patient **can** have both "Cough" and "Fever" → those are independent (not mutually exclusive).
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✓ Quick Recap

- **Mutually exclusive events:** cannot happen together ($P(A \cap B) = 0$).
 - Probability rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - Different from **independence** (where events *can* co-exist but don't affect each other).
 - Important for understanding when features in a model **can overlap** and when they are **strictly separated**.
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